CHANG-SHENG LEE

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EDUCATION

University of California, San Diego Expected Jun. 2028

Master of Science in Computer Science and Engineering

National Taiwan University of Science and Technology Sep. 2021 – Jun. 2025

Bachelor of Science in Computer Science and Information Engineering **GPA 4.19/4.30**

- Certificate of Achievement (Top 5% of class): Fall 2021, Fall 2022, Fall 2023, Spring 2024

WORK EXPERIENCE

Academia Sinica, Institute of Information Science Taipei, Taiwan

Research Assistant Jan. 2025 – Aug. 2026

- Improved LLM-based firewall policy anomaly-detection accuracy from 36% to 99% using SFT and RL
- Built an automated evaluation pipeline for 4 state-of-the-art LLMs across 200 datasets
- Developed an AI agent that translates network intents into firewall rules and regenerates conflict-free policy sets
- Designed a normalization framework reducing firewall maintenance cost by eliminating all rule conflicts
- Selected Publication
  - **Chang-Sheng Lee**, Ling-Jyh Chen. *Firewall Normal Form: A Normalization Framework for Firewall Management and Anomaly Correction*. International Conference on Advanced Network Technologies and Applications, 2026
  - **Chang-Sheng Lee**, I-Chen Lee, Ling-Jyh Chen. *Enhancing Firewall Rule Anomaly Detection via LLM Alignment*. International Conference on Technologies and Applications of Artificial Intelligence, 2025
  - **Chang-Sheng Lee**, Ling-Jyh Chen. *Can AI Outsmart Firewall Errors? a Study on LLMs for Anomaly Generation and Detection*. IEEE Conference on Dependable and Secure Computing, 2025

Research Assistant Intern Jul. 2023 – Jun. 2024

- Uncovered Meshtastic packet discrepancies by decoding SDR-captured LoRa signals and decrypting payloads.
- Corrected official packet documentation with experimental evidence of specification discrepancies
- Pioneered open-source and resilient solutions for disaster recovery and off-grid communication

NTUST GAME Lab (Coop. with International Games System Co., Ltd) Taipei, Taiwan

Software Engineer / Research Assistant May 2022 – Feb. 2024

- Led a team of five to build a rule-based AI bot for real-time Mahjong decision-making and improve win rate by 20%
- Improved AI bot search efficiency for candidate hand states using a Levenshtein distance-based structure
- Developed backend services and automated testing tools supporting online gameplay for 1,000+ players.

PROJECTS

Editable Facial Animation from Image and Reference Video

- Enabled editable facial motion transfer from reference videos to static images with export to Spine2D projects
- Integrated neural landmark detection and generative animation to create editable skeletons, meshes, and weights

SKILLS

Programming Language: C/C++, Python, Java, JavaScript, HTML, CSS, SQL

Tools/Framework: Git, Docker, Linux, Google Cloud Platform, React / Full-Stack Web

ML/LLM: LoRA / PEFT Fine-Tuning, Supervised Fine-Tuning, Reinforcement Learning, LLM agents, Hugging Face, PyTorch